

CORE PURE / COMPLEX NUMBERS

Complex Series

Worked solutions

7 questions · 64 marks

Questions

Question 1	8 marks	2
Question 2	12 marks	4
Question 3	12 marks	6
Question 4	8 marks	8
Question 5	9 marks	10
Question 6	8 marks	12
Question 7	7 marks	14

1. The infinite series C and S are defined by

$$C = \cos 2\theta - \frac{1}{3} \cos 5\theta + \frac{1}{9} \cos 8\theta - \frac{1}{27} \cos 11\theta + \dots$$

$$S = \sin 2\theta - \frac{1}{3} \sin 5\theta + \frac{1}{9} \sin 8\theta - \frac{1}{27} \sin 11\theta + \dots$$

Given that the series C and S are both convergent.

(a) Show that

$$C + iS = \frac{3e^{2i\theta}}{3 + e^{3i\theta}} \quad [4]$$

(b) Hence show that

$$S = \frac{9 \sin 2\theta - 3 \sin \theta}{10 + 6 \cos 3\theta} \quad [4]$$

Solution

(a) Using Euler's formula,

$$\cos x + i \sin x = e^{ix}$$

So

$$\begin{aligned} C + iS &= (\cos 2\theta + i \sin 2\theta) - \frac{1}{3} (\cos 5\theta + i \sin 5\theta) + \frac{1}{9} (\cos 8\theta + i \sin 8\theta) - \dots \\ &= e^{2i\theta} - \frac{1}{3} e^{5i\theta} + \frac{1}{9} e^{8i\theta} - \frac{1}{27} e^{11i\theta} + \dots \end{aligned}$$

Factor out $e^{2i\theta}$:

$$C + iS = e^{2i\theta} \left(1 - \frac{1}{3} e^{3i\theta} + \frac{1}{9} e^{6i\theta} - \frac{1}{27} e^{9i\theta} + \dots \right)$$

This is a geometric series with first term 1 and common ratio

$$r = -\frac{e^{3i\theta}}{3}$$

Since

$$|r| = \frac{|e^{3i\theta}|}{3} = \frac{1}{3} < 1$$

the series converges, so

$$\begin{aligned} C + iS &= e^{2i\theta} \cdot \frac{1}{1 - (-e^{3i\theta}/3)} \\ &= e^{2i\theta} \cdot \frac{1}{1 + e^{3i\theta}/3} \\ &= \frac{3e^{2i\theta}}{3 + e^{3i\theta}} \end{aligned}$$

(b) From part (a),

$$C + iS = \frac{3e^{2i\theta}}{3 + e^{3i\theta}}$$

Multiply numerator and denominator by $3 + e^{-3i\theta}$:

$$\begin{aligned} C + iS &= \frac{3e^{2i\theta}(3 + e^{-3i\theta})}{(3 + e^{3i\theta})(3 + e^{-3i\theta})} \\ &= \frac{9e^{2i\theta} + 3e^{-i\theta}}{9 + 3e^{3i\theta} + 3e^{-3i\theta} + e^{3i\theta}e^{-3i\theta}} \\ &= \frac{9e^{2i\theta} + 3e^{-i\theta}}{10 + 3(e^{3i\theta} + e^{-3i\theta})} \end{aligned}$$

Now use

$$e^{3i\theta} + e^{-3i\theta} = 2 \cos 3\theta$$

so the denominator becomes

$$10 + 6 \cos 3\theta$$

Therefore

$$C + iS = \frac{9e^{2i\theta} + 3e^{-i\theta}}{10 + 6 \cos 3\theta}$$

The denominator is real, so taking imaginary parts gives

$$\begin{aligned} S &= \operatorname{Im} \left[\frac{9e^{2i\theta} + 3e^{-i\theta}}{10 + 6 \cos 3\theta} \right] \\ &= \frac{9 \sin 2\theta + 3(-\sin \theta)}{10 + 6 \cos 3\theta} \end{aligned}$$

Hence

$$S = \frac{9 \sin 2\theta - 3 \sin \theta}{10 + 6 \cos 3\theta}$$

2. (i) Show that

$$(1 - 2e^{2i\theta})(1 - 2e^{-2i\theta}) = 5 - 4\cos 2\theta \quad [3]$$

For a positive integer n , series C and S are defined by

$$C = \cos \theta + 2\cos 3\theta + 4\cos 5\theta + \cdots + 2^{n-1}\cos((2n-1)\theta)$$

$$S = \sin \theta + 2\sin 3\theta + 4\sin 5\theta + \cdots + 2^{n-1}\sin((2n-1)\theta)$$

(ii) Show that

$$C = \frac{2^{n+1}\cos((2n-1)\theta) - 2^n\cos((2n+1)\theta) - \cos \theta}{5 - 4\cos 2\theta}$$

and find a similar expression for S .

[9]

Solution

(i) Expanding the product,

$$\begin{aligned} (1 - 2e^{2i\theta})(1 - 2e^{-2i\theta}) &= 1 - 2e^{2i\theta} - 2e^{-2i\theta} + 4 \\ &= 5 - 2(e^{2i\theta} + e^{-2i\theta}) \end{aligned}$$

Using

$$e^{2i\theta} + e^{-2i\theta} = 2\cos 2\theta$$

gives

$$\begin{aligned} (1 - 2e^{2i\theta})(1 - 2e^{-2i\theta}) &= 5 - 2(2\cos 2\theta) \\ &= 5 - 4\cos 2\theta \end{aligned}$$

as required.

(ii) Let

$$Z = C + iS$$

Then, using $\cos x + i\sin x = e^{ix}$,

$$\begin{aligned} Z &= (\cos \theta + i\sin \theta) + 2(\cos 3\theta + i\sin 3\theta) + \cdots + 2^{n-1}(\cos((2n-1)\theta) + i\sin((2n-1)\theta)) \\ &= e^{i\theta} + 2e^{3i\theta} + 4e^{5i\theta} + \cdots + 2^{n-1}e^{(2n-1)i\theta} \\ &= e^{i\theta}(1 + 2e^{2i\theta} + (2e^{2i\theta})^2 + \cdots + (2e^{2i\theta})^{n-1}) \end{aligned}$$

This is a geometric series with first term 1 and common ratio $2e^{2i\theta}$, so

$$Z = e^{i\theta} \frac{1 - (2e^{2i\theta})^n}{1 - 2e^{2i\theta}} = e^{i\theta} \frac{1 - 2^n e^{2ni\theta}}{1 - 2e^{2i\theta}}$$

Multiply numerator and denominator by $1 - 2e^{-2i\theta}$:

$$\begin{aligned} Z &= e^{i\theta} \frac{1 - 2^n e^{2ni\theta}}{1 - 2e^{2i\theta}} \cdot \frac{1 - 2e^{-2i\theta}}{1 - 2e^{-2i\theta}} \\ &= \frac{e^{i\theta}(1 - 2^n e^{2ni\theta})(1 - 2e^{-2i\theta})}{(1 - 2e^{2i\theta})(1 - 2e^{-2i\theta})} \end{aligned}$$

Using part (i), the denominator is $5 - 4\cos 2\theta$, so

$$\begin{aligned} Z &= \frac{e^{i\theta}(1 - 2^n e^{2ni\theta})(1 - 2e^{-2i\theta})}{5 - 4\cos 2\theta} \\ &= \frac{e^{i\theta} - 2e^{-i\theta} - 2^n e^{i(2n+1)\theta} + 2^{n+1} e^{i(2n-1)\theta}}{5 - 4\cos 2\theta} \end{aligned}$$

Now simplify the first two terms:

$$\begin{aligned}e^{i\theta} - 2e^{-i\theta} &= (\cos \theta + i \sin \theta) - 2(\cos \theta - i \sin \theta) \\&= -\cos \theta + 3i \sin \theta\end{aligned}$$

Hence

$$Z = \frac{-\cos \theta + 3i \sin \theta - 2^n e^{i(2n+1)\theta} + 2^{n+1} e^{i(2n-1)\theta}}{5 - 4 \cos 2\theta}$$

Writing the exponential terms in sine and cosine form,

$$\begin{aligned}Z &= \frac{-\cos \theta - 2^n \cos((2n+1)\theta) + 2^{n+1} \cos((2n-1)\theta)}{5 - 4 \cos 2\theta} \\&\quad + i \frac{3 \sin \theta - 2^n \sin((2n+1)\theta) + 2^{n+1} \sin((2n-1)\theta)}{5 - 4 \cos 2\theta}\end{aligned}$$

Since $Z = C + iS$, equating real and imaginary parts gives

$$C = \frac{2^{n+1} \cos((2n-1)\theta) - 2^n \cos((2n+1)\theta) - \cos \theta}{5 - 4 \cos 2\theta}$$

and

$$S = \frac{2^{n+1} \sin((2n-1)\theta) - 2^n \sin((2n+1)\theta) + 3 \sin \theta}{5 - 4 \cos 2\theta}$$

3. (i) Show that

$$1 - e^{i2\theta} = 2 \sin \theta (\sin \theta - i \cos \theta) \quad [3]$$

(ii) For a positive integer n , the series C and S are defined as follows.

$$\begin{aligned} C &= 1 - \binom{n}{1} \cos 2\theta + \binom{n}{2} \cos 4\theta - \dots + (-1)^n \cos 2n\theta \\ S &= -\binom{n}{1} \sin 2\theta + \binom{n}{2} \sin 4\theta - \dots + (-1)^n \sin 2n\theta \end{aligned}$$

By considering $C + iS$, show that

$$C = 2^n \sin^n \theta \cos \left(n\theta - \frac{n\pi}{2} \right)$$

and find a corresponding expression for S .

[9]

Solution

(i) Using Euler's formula $e^{i2\theta} = \cos 2\theta + i \sin 2\theta$,

$$\begin{aligned} 1 - e^{i2\theta} &= 1 - \cos 2\theta - i \sin 2\theta \\ &= (1 - \cos 2\theta) - i \sin 2\theta \\ &= 2 \sin^2 \theta - i(2 \sin \theta \cos \theta) \\ &= 2 \sin \theta (\sin \theta - i \cos \theta) \end{aligned}$$

(ii) Combine the two series into one complex expression:

$$\begin{aligned} C + iS &= 1 - \binom{n}{1} (\cos 2\theta + i \sin 2\theta) + \binom{n}{2} (\cos 4\theta + i \sin 4\theta) - \dots + (-1)^n (\cos 2n\theta + i \sin 2n\theta) \\ &= 1 - \binom{n}{1} e^{i2\theta} + \binom{n}{2} e^{i4\theta} - \dots + (-1)^n e^{i2n\theta} \\ &= \sum_{r=0}^n \binom{n}{r} (-1)^r e^{i2r\theta} \\ &= \sum_{r=0}^n \binom{n}{r} (-e^{i2\theta})^r \\ &= (1 - e^{i2\theta})^n \end{aligned}$$

by the binomial theorem.

From part (i),

$$1 - e^{i2\theta} = 2 \sin \theta (\sin \theta - i \cos \theta)$$

so

$$\begin{aligned} C + iS &= [2 \sin \theta (\sin \theta - i \cos \theta)]^n \\ &= 2^n \sin^n \theta (\sin \theta - i \cos \theta)^n \end{aligned}$$

Now factor out $-i$:

$$\sin \theta - i \cos \theta = -i(\cos \theta + i \sin \theta)$$

Since

$$-i = e^{-i\pi/2}$$

this becomes

$$\sin \theta - i \cos \theta = e^{-i\pi/2} (\cos \theta + i \sin \theta)$$

Therefore, by De Moivre's theorem,

$$\begin{aligned}(\sin \theta - i \cos \theta)^n &= \left(e^{-i\pi/2} (\cos \theta + i \sin \theta) \right)^n \\&= e^{-in\pi/2} e^{in\theta} \\&= \cos \left(n\theta - \frac{n\pi}{2} \right) + i \sin \left(n\theta - \frac{n\pi}{2} \right)\end{aligned}$$

Substituting this into $C + iS$,

$$C + iS = 2^n \sin^n \theta \left(\cos \left(n\theta - \frac{n\pi}{2} \right) + i \sin \left(n\theta - \frac{n\pi}{2} \right) \right)$$

Equating real and imaginary parts gives

$$C = 2^n \sin^n \theta \cos \left(n\theta - \frac{n\pi}{2} \right)$$

and

$$S = 2^n \sin^n \theta \sin \left(n\theta - \frac{n\pi}{2} \right)$$

4. The infinite series C and S are defined as follows:

$$C = \cos \alpha + r \cos(\alpha + \beta) + r^2 \cos(\alpha + 2\beta) + \dots$$

$$S = \sin \alpha + r \sin(\alpha + \beta) + r^2 \sin(\alpha + 2\beta) + \dots$$

where r is a real number and $|r| < 1$.

By considering $C + iS$, show that

$$C = \frac{\cos \alpha - r \cos(\alpha - \beta)}{1 + r^2 - 2r \cos \beta}$$

and find a corresponding expression for S .

[8]

Solution

Let

$$Z = C + iS$$

Then

$$\begin{aligned} Z &= (\cos \alpha + r \cos(\alpha + \beta) + r^2 \cos(\alpha + 2\beta) + \dots) \\ &\quad + i (\sin \alpha + r \sin(\alpha + \beta) + r^2 \sin(\alpha + 2\beta) + \dots) \\ &= \sum_{n=0}^{\infty} r^n (\cos(\alpha + n\beta) + i \sin(\alpha + n\beta)) \end{aligned}$$

Using Euler's formula,

$$\cos \theta + i \sin \theta = e^{i\theta}$$

so

$$\begin{aligned} Z &= \sum_{n=0}^{\infty} r^n e^{i(\alpha + n\beta)} \\ &= e^{i\alpha} \sum_{n=0}^{\infty} (re^{i\beta})^n \end{aligned}$$

Since

$$|re^{i\beta}| = |r| < 1$$

this is a convergent geometric series, so

$$\begin{aligned} Z &= e^{i\alpha} \cdot \frac{1}{1 - re^{i\beta}} \\ &= \frac{e^{i\alpha}}{1 - re^{i\beta}} \end{aligned}$$

Now multiply top and bottom by the conjugate-like factor $1 - re^{-i\beta}$:

$$Z = \frac{e^{i\alpha}(1 - re^{-i\beta})}{(1 - re^{i\beta})(1 - re^{-i\beta})}$$

First simplify the denominator:

$$\begin{aligned} (1 - re^{i\beta})(1 - re^{-i\beta}) &= 1 - r(e^{i\beta} + e^{-i\beta}) + r^2 \\ &= 1 - r(2 \cos \beta) + r^2 \\ &= 1 + r^2 - 2r \cos \beta \end{aligned}$$

Now simplify the numerator:

$$\begin{aligned}e^{i\alpha}(1 - re^{-i\beta}) &= e^{i\alpha} - re^{i\alpha}e^{-i\beta} \\ &= e^{i\alpha} - re^{i(\alpha-\beta)}\end{aligned}$$

Hence

$$Z = \frac{e^{i\alpha} - re^{i(\alpha-\beta)}}{1 + r^2 - 2r \cos \beta}$$

Write the exponentials in terms of sine and cosine:

$$\begin{aligned}Z &= \frac{\cos \alpha + i \sin \alpha - r(\cos(\alpha - \beta) + i \sin(\alpha - \beta))}{1 + r^2 - 2r \cos \beta} \\ &= \frac{(\cos \alpha - r \cos(\alpha - \beta)) + i(\sin \alpha - r \sin(\alpha - \beta))}{1 + r^2 - 2r \cos \beta}\end{aligned}$$

But $Z = C + iS$, so equating real and imaginary parts gives

$$C = \frac{\cos \alpha - r \cos(\alpha - \beta)}{1 + r^2 - 2r \cos \beta}$$

and

$$S = \frac{\sin \alpha - r \sin(\alpha - \beta)}{1 + r^2 - 2r \cos \beta}$$

Therefore,

$$C = \frac{\cos \alpha - r \cos(\alpha - \beta)}{1 + r^2 - 2r \cos \beta}, \quad S = \frac{\sin \alpha - r \sin(\alpha - \beta)}{1 + r^2 - 2r \cos \beta}$$

5. (i) For values of θ for which the expressions are defined, show that

$$1 + i \tan \theta = \sec \theta (\cos \theta + i \sin \theta) \quad [2]$$

- (ii) For a positive integer n , the series C and S are defined by

$$C = 1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots$$

$$S = \binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \binom{n}{5} \tan^5 \theta - \dots$$

where in each case the pattern continues until the power of $\tan \theta$ is at most n .

By considering $C + iS$, show that

$$C = \sec^n \theta \cos n\theta$$

and find a corresponding expression for S .

[7]

Solution

- (i) For the expressions to be defined, we need

$$\cos \theta \neq 0$$

Now

$$\begin{aligned} \sec \theta (\cos \theta + i \sin \theta) &= \sec \theta \cos \theta + i \sec \theta \sin \theta \\ &= 1 + i \tan \theta \end{aligned}$$

as required.

- (ii) Consider $C + iS$.

Using the given definitions,

$$C + iS = \left(1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots \right) + i \left(\binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \binom{n}{5} \tan^5 \theta - \dots \right)$$

Now note that

$$(i \tan \theta)^0 = 1, \quad (i \tan \theta)^1 = i \tan \theta, \quad (i \tan \theta)^2 = -\tan^2 \theta, \quad (i \tan \theta)^3 = -i \tan^3 \theta$$

so the terms in $C + iS$ match the binomial expansion of $(1 + i \tan \theta)^n$. Hence

$$C + iS = \sum_{k=0}^n \binom{n}{k} (i \tan \theta)^k = (1 + i \tan \theta)^n$$

Using part (i),

$$1 + i \tan \theta = \sec \theta (\cos \theta + i \sin \theta)$$

Therefore

$$\begin{aligned} C + iS &= (1 + i \tan \theta)^n \\ &= [\sec \theta (\cos \theta + i \sin \theta)]^n \\ &= \sec^n \theta (\cos \theta + i \sin \theta)^n \end{aligned}$$

By de Moivre's theorem,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

So

$$C + iS = \sec^n \theta (\cos n\theta + i \sin n\theta)$$

Equating real and imaginary parts gives

$$C = \sec^n \theta \cos n\theta$$

and

$$S = \sec^n \theta \sin n\theta$$

6. The convergent series C and S are defined by

$$C = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n \cos((4n+1)\theta)$$

$$S = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n \sin((4n+1)\theta)$$

(a) Show that

$$C + iS = \frac{2e^{i\theta}}{2 + e^{4i\theta}} \quad [4]$$

(b) Hence show that

$$C = \frac{4 \cos \theta + 2 \cos 3\theta}{5 + 4 \cos 4\theta} \quad [4]$$

Solution

(a) Using Euler's formula,

$$\cos x + i \sin x = e^{ix}$$

so

$$\begin{aligned} C + iS &= \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n (\cos((4n+1)\theta) + i \sin((4n+1)\theta)) \\ &= \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n e^{i(4n+1)\theta} \\ &= e^{i\theta} \sum_{n=0}^{\infty} \left(-\frac{e^{4i\theta}}{2}\right)^n \end{aligned}$$

This is a geometric series with common ratio

$$r = -\frac{e^{4i\theta}}{2}$$

and

$$|r| = \frac{1}{2} < 1$$

so it converges and its sum is

$$\begin{aligned} C + iS &= e^{i\theta} \cdot \frac{1}{1 - r} \\ &= e^{i\theta} \cdot \frac{1}{1 + \frac{1}{2}e^{4i\theta}} \\ &= \frac{2e^{i\theta}}{2 + e^{4i\theta}} \end{aligned}$$

(b) Starting from part (a),

$$C + iS = \frac{2e^{i\theta}}{2 + e^{4i\theta}}$$

Multiply top and bottom by $2 + e^{-4i\theta}$:

$$\begin{aligned} C + iS &= \frac{2e^{i\theta}(2 + e^{-4i\theta})}{(2 + e^{4i\theta})(2 + e^{-4i\theta})} \\ &= \frac{4e^{i\theta} + 2e^{-3i\theta}}{(2 + e^{4i\theta})(2 + e^{-4i\theta})} \end{aligned}$$

Now simplify the denominator:

$$\begin{aligned} (2 + e^{4i\theta})(2 + e^{-4i\theta}) &= 4 + 2e^{4i\theta} + 2e^{-4i\theta} + 1 \\ &= 5 + 2(e^{4i\theta} + e^{-4i\theta}) \\ &= 5 + 4\cos 4\theta \end{aligned}$$

since $e^{ix} + e^{-ix} = 2\cos x$.

Therefore

$$C + iS = \frac{4e^{i\theta} + 2e^{-3i\theta}}{5 + 4\cos 4\theta}$$

The denominator is real, so taking real parts gives

$$\begin{aligned} C &= \frac{4\cos\theta + 2\cos(-3\theta)}{5 + 4\cos 4\theta} \\ &= \frac{4\cos\theta + 2\cos 3\theta}{5 + 4\cos 4\theta} \end{aligned}$$

7. Let

$$C = \sum_{r=0}^{n-1} \cos^2 \left(\theta + \frac{2\pi r}{n} \right)$$

$$S = \sum_{r=0}^{n-1} \sin \left(\theta + \frac{2\pi r}{n} \right) \cos \left(\theta + \frac{2\pi r}{n} \right)$$

where n is an integer greater than 2. Let $\exp(x)$ denote e^x .

By considering the real and imaginary parts of

$$\sum_{r=0}^{n-1} \exp \left(2i \left(\theta + \frac{2\pi r}{n} \right) \right)$$

show that $C = \frac{n}{2}$ and $S = 0$.

[7]

Solution

Let

$$Z = \sum_{r=0}^{n-1} \exp \left(2i \left(\theta + \frac{2\pi r}{n} \right) \right)$$

Using $\exp(ix) = \cos x + i \sin x$, this becomes

$$Z = \sum_{r=0}^{n-1} \left(\cos \left(2\theta + \frac{4\pi r}{n} \right) + i \sin \left(2\theta + \frac{4\pi r}{n} \right) \right)$$

So

$$\operatorname{Re}(Z) = \sum_{r=0}^{n-1} \cos \left(2\theta + \frac{4\pi r}{n} \right)$$

$$\operatorname{Im}(Z) = \sum_{r=0}^{n-1} \sin \left(2\theta + \frac{4\pi r}{n} \right)$$

Now rewrite Z as a geometric series:

$$Z = \sum_{r=0}^{n-1} \exp \left(2i\theta + \frac{4\pi ir}{n} \right)$$

$$= \exp(2i\theta) \sum_{r=0}^{n-1} \left(\exp \left(\frac{4\pi i}{n} \right) \right)^r$$

Let

$$\omega = \exp \left(\frac{4\pi i}{n} \right)$$

Then

$$Z = \exp(2i\theta) \sum_{r=0}^{n-1} \omega^r$$

Since $n > 2$, we have $0 < \frac{2}{n} < 1$, so $\omega \neq 1$. Therefore the geometric series formula gives

$$Z = \exp(2i\theta) \frac{1 - \omega^n}{1 - \omega}$$

But

$$\begin{aligned}\omega^n &= \left(\exp \left(\frac{4\pi i}{n} \right) \right)^n \\ &= \exp(4\pi i) \\ &= 1\end{aligned}$$

So

$$\begin{aligned}Z &= \exp(2i\theta) \frac{1-1}{1-\omega} \\ &= 0\end{aligned}$$

Hence both the real and imaginary parts of Z are zero:

$$\begin{aligned}\sum_{r=0}^{n-1} \cos \left(2\theta + \frac{4\pi r}{n} \right) &= 0 \\ \sum_{r=0}^{n-1} \sin \left(2\theta + \frac{4\pi r}{n} \right) &= 0\end{aligned}$$

Now use the double-angle identities.

For C ,

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

with $x = \theta + \frac{2\pi r}{n}$. Then

$$\begin{aligned}C &= \sum_{r=0}^{n-1} \cos^2 \left(\theta + \frac{2\pi r}{n} \right) \\ &= \sum_{r=0}^{n-1} \frac{1 + \cos \left(2\theta + \frac{4\pi r}{n} \right)}{2} \\ &= \frac{1}{2} \sum_{r=0}^{n-1} 1 + \frac{1}{2} \sum_{r=0}^{n-1} \cos \left(2\theta + \frac{4\pi r}{n} \right) \\ &= \frac{1}{2}(n) + \frac{1}{2}(0) = \frac{n}{2}\end{aligned}$$

For S ,

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

so, again with $x = \theta + \frac{2\pi r}{n}$,

$$\begin{aligned}S &= \sum_{r=0}^{n-1} \sin \left(\theta + \frac{2\pi r}{n} \right) \cos \left(\theta + \frac{2\pi r}{n} \right) \\ &= \frac{1}{2} \sum_{r=0}^{n-1} \sin \left(2\theta + \frac{4\pi r}{n} \right) \\ &= \frac{1}{2}(0) = 0\end{aligned}$$

Therefore

$$C = \frac{n}{2} \quad \text{and} \quad S = 0$$